Coordinate-Governed Mapping of Source-Local Scientific Projections: A Fixed-Panel Study

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Abstract

Background. Scientific statements can appear to agree after their polarity, measurement, context or provenance has been discarded. We test a small decision-layer rule that keeps those coordinates explicit when mapping already-structured scientific projections.

Methods. An initial 32-case panel motivated a prospectively frozen, case-identifier-disjoint 32-case holdout spanning three authored case families. We compared coordinate-governed mapping with flat predicate canonicalization and an exact-coordinate conservative rule. Bootstrap intervals are descriptive diagnostics for this fixed panel, not population uncertainty. A separate deterministic conformance battery contains 400 generated records instantiating eight unique decision archetypes across five labels and ten identifier variants.

Results. On the holdout, coordinate-governed mapping made no false merges, whereas flat canonicalization made six of 32 (difference -0.1875; fixed-panel 95% bootstrap diagnostic $[-0.34375, -0.0625]$). All six errors were polarity contrasts; the other two families (19 cases) did not discriminate the rules. The false-split difference from the conservative rule was zero. In the conformance battery, the complete rule scored 400/400, the strong semantic product 250/400, and canonical matching 50/400. An information-equivalent typed implementation also scored 400/400.

Conclusions. These panels support a bounded observation: the complete polarity-sensitive non-merge condition avoids the six false agreements created by the registered flat rule. They do not establish population error rates, superiority over deployed integration systems, the independent value of every coordinate, raw-text extraction performance or downstream scientific utility.

Keywords: semantic interoperability; knowledge representation; evidence provenance; coordinate systems; scientific mapping.

1 Introduction

Scientific claims are distributed across fields that use different terms, measurements, contexts and representational conventions. Identical predicates can refer to different objects, and a positive claim from one source can be mistakenly merged with a negative claim from another. This report studies only the downstream decision of whether two already-structured scientific projections may be treated as one statement. It does not study retrieval or extraction from raw text.

Scientific information extraction, schema induction and knowledge-graph construction supply increasingly structured upstream representations [2, 16, 4, 8, 9]. Ontology matching studies equivalence, subsumption and disjointness [6]; alignment repair also shows that individually plausible correspondences can make a combined ontology incoherent [12]. Ontological unpacking, I-ADOPT and FAIR 2.0 make scientific-variable structure and mapping conditions explicit [1, 11, 14]. Provenance-bearing statements and target-relative stance further motivate preserving source and polarity rather than collapsing them into a scalar match [13, 7]. The present contribution is smaller than these extraction, matching and modelling programmes: it tests a claim-relative merge rule only after the source projections have already been constructed.

The rule records referent, construct, measurement, context, polarity, modality and attribution separately, permits an explicit non-merge decision, and leaves insufficiently supported relations undetermined. We evaluate the rule on a prospectively frozen holdout and report a separate deterministic contract test. This evidence architecture supports a Brief Report: the scientific units are three confirmatory case families and eight unique conformance archetypes; the 432 scored rows are not independent scientific observations.

The individual coordinates, inconsistency-sensitive separation, provenance and abstention are inherited from the neighbouring traditions cited above. The increment tested here is their frozen, claim-relative composition into one decision condition, its prospective fixed-panel evaluation and its executable contract. We do not claim a new general ontology language or a new extraction method.

2 Methods

2.1 Mapping decisions

For a source record s, the structured projection is

$$P_s = \langle R_s, C_s, M_s, X_s, \rho_s, \mu_s, a_s, \delta_s, \Pi_s \rangle, \tag{1}$$

where the entries denote referent, construct, measurement, context, polarity, modality, attribution, discourse relation and assumptions or missing information. Table 1 gives the frozen precedence rule. Each condition is evaluated from top to bottom and the first applicable row is returned. Equality means exact equality of the exposed identifiers or enum value. An empty coordinate is not declared incompatible; absence produces an undetermined decision only when it is recorded as an unresolved ambiguity or prevents predicate normalization. The study does not infer absent coordinates from source text.

First applicable condition	**Fine relation**	**Decision**
Either record retains an unresolved ambiguity, or predicates are not normalized to the same relation	unresolved	undetermined
Both nonempty referent identifiers differ	distinct referent	typed non-merge
Both nonempty construct identifiers differ	distinct construct	typed non-merge
Both nonempty measurement identifiers differ	distinct measurement	typed non-merge
Both nonempty temporal, attribution, discourse or assumption coordinates differ, or modalities differ	contextual difference	typed non-merge
Known polarities differ; the fine relation is contradiction when both modalities are asserted and contextual difference otherwise	contradiction or contextual difference	typed non-merge
None of the preceding conditions applies	compatible	admissible merge

Table 1: Normative holdout decision rule. For example, aligned asserted projections with opposite polarity return contradiction and hence typed non-merge; an explicit unresolved ambiguity returns undetermined.

2.2 Fixed panels and scientific units

The development panel and the confirmatory holdout each contain 32 already-structured cases assembled from pinned public resources and deterministic standards. The holdout was selected by an outcome-blind round-robin window and shares no case identifiers with development, although two combinations of source locator and content hash recur. Before holdout outputs were examined, the protocol fixed the case digest, source revisions, evaluator identities, bootstrap seed, margins and terminal rule.

The holdout contains three authored families: 13 different-name/same-referent cases, 13 polarity/modality/attribution/context cases, and six valid/invalid representation-mapping cases. A case is the scoring unit for the registered rates and paired differences. It is not an independent population or cross-domain generalization unit. The three families are the appropriate scale for assessing scientific breadth, and only the second family contains observed between-rule discrimination. The expected decisions, candidate rule and evaluation were produced within the same authorship and research programme. Prospective freezing limits outcome-contingent adaptation but does not turn same-programme gold into independent adjudication.

2.3 Comparators and analysis

All rules received the same structured inputs. Flat predicate canonicalization treated compatible predicates as sufficient merge evidence and was the predeclared false-merge comparator. It is intentionally weak and isolates the consequence of removing the registered coordinate condition. The exact-coordinate conservative rule could abstain and served as the false-split control. These rules are not implementations of complete contemporary ontology-matching or integration systems.

All rates use the 32 cases as denominator. A false merge is a compatible prediction when gold is any typed non-merge relation; a false split is a typed non-merge prediction when gold is compatible. An unresolved prediction counts as abstention, not as either error. Thus abstaining on a gold non-merge case can lower exact-match accuracy without creating a false split. The flat comparator returns compatible for equal predicates and unresolved otherwise. The conservative comparator returns compatible only when every exposed coordinate is exactly equal and unresolved otherwise. Registered ablations remove only the named coordinate comparison from the ordered rule above; the force-compatibility ablation maps every typed non-merge relation to compatible.

The confirmatory gate had two non-compensatory conditions. The coordinate-governed-minus-flat false-merge estimate had to be at most -0.05 with the upper bound of its paired fixed-panel bootstrap diagnostic below zero. The coordinate-governed-minus-conservative false-split estimate had to be at most $+0.03$ with the diagnostic upper bound no greater than $+0.03$. Paired diagnostics use 10,000 percentile-bootstrap resamples with seed 20260817. The rules are deterministic, so stochastic repetitions were not manufactured. The development and confirmatory panels are not pooled.

2.4 Deterministic contract conformance

A separately frozen generator produced 400 records as five domain labels $\times$ eight decision archetypes $\times$ ten identifier variants. Domain and variant change identifiers and commitments but not the candidate-visible decision state. The effective mechanism census is therefore eight unique states, each repeated 50 times. This battery tests deterministic interface and specification conformance; it does not supply 400 independent observations.

The complete rule can return merge, obstruction, plural view or unresolved. The strong semantic product omits the provenance/plural-view and global-cycle relations and cannot emit plural view;

its 250/400 score is therefore an attainability result under that declared interface, not evidence of general capability superiority. Canonical matching uses a binary threshold. The fourth comparator is definitionally information-equivalent to the complete rule and must tie it exactly.

This four-way contract refines, rather than duplicates, the holdout alphabet: merge maps to admissible merge; obstruction and plural view both map to typed non-merge, with plural view retaining both source projections; and unresolved maps to undetermined. The mapping is therefore many-to-one. The eight archetypes contain one merge, four obstruction, two plural-view and one unresolved state, each repeated across labels and identifiers. The reported conformance scores are exact four-way scores computed before this crosswalk; they are not a second evaluation of the holdout's fine-grained relation labels.

3 Results

On the development panel, coordinate-governed mapping made no false merges or false splits; flat canonicalization made four false merges. This outcome motivated the frozen holdout and is not pooled into its verdict.

Rule	**Acc.**	**False merge**	**False split**	**Abstain**
Coordinate-governed	1.0000	0.0000	0.0000	0.0000
Flat canonicalization	0.8125	0.1875	0.0000	0.0000
Exact conservative	0.8125	0.0000	0.0000	0.1875

Table 2: Outcomes on the 32-case confirmatory holdout. Rates use one decision per case. The coordinate-governed-minus-flat false-merge difference is -0.1875 (fixed-panel 95% bootstrap diagnostic $[-0.34375, -0.0625]$); the false-split difference from the conservative rule is zero, with diagnostic $[0, 0]$.

All six flat-rule false merges were polarity contrasts within the family covering polarity, modality, attribution and context. Every rule was correct in the other two families (19 cases). Forcing compatibility without the non-merge terminal increased the false-merge rate by 0.1875; removing the modality, polarity, attribution and discourse group produced the same change. Ablations of referent, construct, measurement and temporal context produced zero measured change on this panel. Those nulls indicate absent comparison opportunity, not dispensability.

The preserved null-and-baseline battery makes both degeneracies explicit, and we state them because they bound the headline. Within compared pairs, ten of the eleven comparison coordinates tracked by the battery (all but polarity, including the predicate itself) never differ on either panel, and a minimal rule restricted to predicate, modality and polarity reproduces every decision of the full mechanism exactly (accuracy 1.0000, no false merges, no false splits, on both 32-case panels). On this evidence the complete coordinate set is sufficient but only its polarity-bearing subset is exercised; the panels do not demonstrate that the remaining coordinates are necessary. The registered flat rule is degenerate in the complementary direction: predicate equality holds in all 32 cases, so it merges every pair, and its false-merge rate is the panel's non-merge base rate (0.1875 on the confirmatory holdout, 0.125 on the initial panel). The headline contrast therefore separates the mechanism from a constant always-merge policy, not from a competitive integrator. As paired significance evidence, exact two-sided McNemar tests give $p = 0.03125$ on the confirmatory holdout (six discordant pairs) and $p = 0.125$ on the initial panel (four discordant pairs); all ten discordant pairs favour the coordinate-governed rule, and no pooled test was computed by protocol.

A standard-library scorer that imports none of the original analysis code reproduced the holdout decisions and verdict from the frozen records. It shares repository snapshot and custody and is therefore an implementation check, not external replication.

Across the 400 generated conformance records, the complete rule scored 400/400, the strong semantic product 250/400, and canonical matching 50/400. The complete-minus-strong-semantic-product difference was +0.375, with a domain-stratified fixed-battery bootstrap diagnostic of $[0.3275, 0.4225]$ and 150–0 discordant decisions. The complete rule made zero false merge decisions and retained all clean merges. The information-equivalent typed comparator scored 400/400 with zero decision mismatches. Because these are repetitions of eight authored states, the counts and diagnostics establish exact conformance to the registered interfaces but do not measure population performance.

4 Discussion

The confirmatory observation is narrow but falsifiable. The complete polarity-sensitive non-merge condition prevented six false agreements produced when the registered flat rule discarded that condition, while the predeclared false-split guard was satisfied. Insufficient evidence remained distinct from a negative decision. The separate conformance battery confirms its own four-way contract exactly and an information-equivalent typed implementation reproduces that contract.

Several limits determine the interpretation. First, all observed discrimination lies in one authored family; the other 19 holdout cases and four coordinate-ablation cells are null, and a three-coordinate restriction reproduces every decision, so coordinate necessity beyond the polarity-bearing subset is untested. Second, the comparator that fails on the holdout was designed to be weak — on these panels it is a constant always-merge classifier whose error rate equals the base rate — and no strong same-universe external integration system was executed. The paired evidence accordingly rests on ten discordant pairs across the two panels. Two established lineages bound what such an external comparison would mean: truth-discovery and conflicting-data integration methods resolve disagreement through source-reliability estimation rather than typed identity coordinates [15, 3, 10], and the OAEI campaigns evaluate schema- and instance-level correspondence systems at scale [5]; no method from either lineage was executed on these panels, so no relative claim against them is made. Third, the holdout is identifier-disjoint but not fully source-disjoint, and its expected decisions and candidate rule remain within one authorship and research programme. Fourth, the 400-record battery contains only eight distinct candidate-visible states, and the strong semantic product has a narrower terminal alphabet. These facts prevent population-like or general-superiority interpretations.

The report also says nothing about recovering the coordinates from raw text, expert agreement over a broader atlas, naturalistic transport, deployed integration, or downstream answer quality. The result is best read as a small methods observation and an executable interface contract. A broader paper would need independently governed, source-disjoint naturalistic cases, strong same-universe comparators at matched information, and evaluation of raw-text or downstream use. Those optional successors are not required for this Brief Report and are not represented as completed evidence.

5 Data and software availability

The frozen case records, source registry, protocols, deterministic analyses and independent implementation check are available in the public ORION repository snapshot at commit `3813e6495`:

https://github.com/SzeChunYiu/ORION/tree/3813e64955f03926b0276e50c21b9a65154ad3db. Within that snapshot, the holdout protocol, cases, frozen analysis and independent scorer are, respectively, `protocol/PUBLIC_REFERENCE_CONFIRMATORY_EXECUTION_V1.json`, `gold/adjudicated/public-reference-v1.1-confirmatory/PUBLIC_REFERENCE_GOLD_V1.jsonl`, `evidence/public-reference-v1.1-confirmatory/CONFIRMATORY_ANALYSIS.json` and `scripts/verify_confirmatory_independent.py`. The normative holdout evaluator is `src/orion/knowledge/semantics.py`; the four-way contract and its independent reconstruction are `research/claim_expansion/p3/P3_X_PROTOCOL_V1.md` and `research/claim_expansion/p3/verify_p3_x_independent.py` at the repository root. Code is licensed under Apache-2.0 and manuscript text under the Creative Commons Attribution 4.0 International licence; upstream resource licences remain controlling. Restricted third-party text is not redistributed. A repository DOI may be added after long-term deposit; none is asserted in this version.

Ethics and consent

Not applicable. This computational study involved no human participants, personal data or animals.

Author contributions

Sze Chun Yiu: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing, visualization and project administration.

Competing interests

No competing interests were disclosed.

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